ATR First-Hour 50% Retracement Screener PRO

A professional Pine Screener whitepaper, implementation guide, and operator tutorial

Document	Whitepaper & Tutorial
Version	1.0
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Platform	TradingView Pine Script v6 / Pine Screener
Primary Market	US equities, regular trading hours

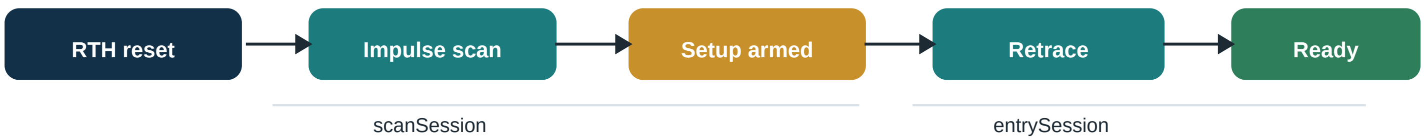
This document explains the design of the upgraded screener, the research basis for the implementation decisions, the screener column contract, and the steps required to deploy and operate the tool in TradingView.

1. Executive Summary

The upgraded script converts the original ATR first-hour retracement screener into a more robust Pine Screener companion. It preserves the original core premise: after a meaningful opening impulse, the symbol becomes actionable when price returns to the selected retracement level, defaulting to 50%.

The primary improvements are engineered around TradingView's documented Pine Screener constraints: the screener scans watchlists through one selected indicator, uses indicator plots and alert conditions as filters/columns, and adds the first ten plots as default columns. The refactor therefore treats those first ten plots as a formal output contract rather than incidental chart plots [1][2].

Key upgrades include robust new-session detection for RTH-only and extended-hours data feeds, optional prior-bar ATR to avoid self-inflating trigger thresholds, timeframe gating for first-hour reliability, quality filters, configurable ready modes, clearer diagnostics, and cleaner alert messages.



2. Research Findings Applied

Finding	Design implication
Pine Screener uses custom indicators to scan watchlists and can sort/analyze indicator plot values [1].	The upgraded script keeps all useful screener outputs as plot() series, mostly hidden from the chart but available to the screener.
When an indicator is selected, Pine Screener adds the first ten plots as columns and adds two alert conditions by default [1].	The first ten plot calls are deliberately ordered: setup flags, ready flags, levels, trigger strength, risk, and distance.
The screener requires at least one plot() or alertcondition() output and only one indicator per screen [1].	The script remains a single indicator with a complete output contract; it does not depend on external indicator-on-indicator inputs.
time() returns a timestamp or na for session membership, and session time zones can be specified [3].	RTH, scan, and entry windows are implemented through time() with explicit America/New_York defaults and selectable session days.
display.none hides plots while preserving their calculated values and supports plot placeholders in alertcondition messages [4].	Screener columns are hidden on the chart but remain usable for scans and alert text.
alertcondition() is indicator-only, works through the Create Alert condition dropdown, and accepts placeholder-based const messages [5][6].	The script provides Long Ready and Short Ready alert conditions with level/target/risk placeholders.

3. Methodology

The screener is not a trading strategy or order execution system. It is a detection layer that identifies symbols exhibiting a specific intraday pattern and emits reference levels for human review.

Opening impulse model

• The regular session open is captured at the first RTH bar.

• During the scan window, the script measures the directional impulse either from the session open or from a single candle body.

• The impulse must reach the configured ATR multiple before a setup is armed.

• The retracement level is measured from the impulse extreme back toward the impulse base. At 50%, a 2xATR impulse implies roughly 1xATR of reference stop distance when the stop is the impulse base.

State machine

• RTH reset: clears prior state and captures dayOpen, with day-change protection for RTH-only data feeds.

• Impulse scan: evaluates the first-hour move against ATR and quality filters.

• Setup armed: stores impulse base, extreme, retracement, stop, target, trigger xATR, and risk xATR.

• Retracement ready: emits Long Ready or Short Ready when the selected entry mode is satisfied.

• Expiration: optionally cancels stale armed setups by bar count or at the end of the entry window.

Important operational assumption

This method is intended for intraday scans. The script defaults to enforcing a maximum 30-minute timeframe because the first-hour scan window and 09:30 open are most reliable on 1m, 5m, 15m, or 30m bars. Pine Screener itself supports a defined set of timeframes, and users must change the screener from its default daily interval to an intraday interval for this tool [1].

4. Quality Improvements Over the Original Script

Area	Upgrade	Why it matters
Session reset	New RTH detection uses both in-session transition and calendar-day change.	Prevents missed resets on RTH-only feeds where the previous bar may also be an RTH bar from the prior trading day.
ATR threshold	Optional prior-bar ATR for trigger calculations.	Avoids allowing a large trigger bar to inflate the threshold it must pass.
Screener contract	First ten plots are intentionally ordered and documented.	Makes filters, sorting, and alert placeholders predictable.
Timeframe control	Intraday plus max-timeframe gate.	Reduces false results from timeframes that do not align well with the first-hour window.
Entry modes	Touch, close at/through, or reclaim close after touch.	Supports both aggressive real-time screens and conservative bar-close workflows.
Risk diagnostics	Trigger xATR, Risk xATR, Distance xATR, ATR %, Open Gap xATR.	Lets the operator sort for quality instead of merely seeing binary setup flags.
Filters	Min price, ATR %, max gap xATR, min/max risk xATR, max age.	Improves practical scan quality for broad US stock watchlists.
Alerts	Placeholders for retrace, stop, target, trigger xATR, and risk xATR.	Creates actionable messages without relying on dynamic alert strings.

5. Pine Screener Column Contract

The first ten plot calls are the primary screener columns. These should be treated as a stable interface: rename or reorder them only if you also update alerts, filters, and operating documentation.

#	Plot title	Type	Meaning	Typical use
1	Long Setup	0/1	Long setup is armed and waiting for retracement.	Filter = 1
2	Short Setup	0/1	Short setup is armed and waiting for retracement.	Filter = 1
3	Long Ready	0/1	Long setup reached the ready condition on this bar.	Alert/filter = 1
4	Short Ready	0/1	Short setup reached the ready condition on this bar.	Alert/filter = 1
5	Retrace Level	Price	Reference entry/retracement level.	Chart review
6	Stop Level	Price	Reference stop at impulse base.	Risk planning
7	Target Level	Price	Reference target at selected R multiple.	Reward planning
8	Trigger xATR	Number	Opening impulse divided by ATR used at trigger.	Sort/filter strength
9	Risk xATR	Number	Stop distance divided by ATR used at trigger.	Filter risk
10	Distance xATR	Number	Current distance from retracement, normalized by ATR.	Sort nearest setups

Secondary hidden diagnostics include ATR Used, Signal Code, Setup Age Bars, Open Gap xATR, and ATR %. TradingView alert placeholders can access the first twenty output series using plot placeholders, while the script uses named plot placeholders in alertcondition messages [6].

6. Inputs and Suggested Presets

Profile	Suggested settings	Use case
Balanced discretionary	5m or 15m; ATR length 14; trigger 2.0x; 50% retracement; prior ATR on; Touch retracement; confirm-on-close off.	Fast intraday screening where the operator reviews the chart before acting.
Conservative	15m or 30m; trigger 2.5–3.0x; Reclaim close after touch; confirm-on-close on; max gap 2–3xATR; max setup age 12–24 bars.	Fewer, more confirmed candidates.
High volatility	Min ATR % > 0; max risk xATR set by plan; Trigger xATR >= 2.5; consider wick extremes only after testing.	Avoids low-volatility symbols and prioritizes stronger movement.

The default settings intentionally remain close to the original script: 14-period ATR, 2xATR trigger, 50% retracement, first-hour scan, and touch-based ready behavior. More conservative behavior is available through inputs rather than forced by default.

7. Tutorial: Deploying in TradingView

Step 1 — Install the script

• Open TradingView Supercharts and then Pine Editor.

• Paste the upgraded .pine source file into the editor.

• Save the script, add it to a chart once, and mark it as a favorite. Pine Screener requires the selected indicator to be available from your favorites list [1].

Step 2 — Open Pine Screener

• Open Products → Screeners → Pine.

• Select the watchlist you want to scan.

• Select the upgraded indicator from the indicator dropdown.

• Change the screener timeframe to 5m, 15m, or 30m. Avoid leaving the indicator on the default daily interval for this intraday method.

Step 3 — Build filters

Objective	Filter / sort action
Find actionable long retracements	Long Ready = 1
Find actionable short retracements	Short Ready = 1
Monitor setups before entry	Long Setup = 1 or Short Setup = 1
Sort setups closest to entry	Sort Distance xATR near zero
Require stronger impulse	Trigger xATR >= 2.5 or your chosen threshold
Avoid wide stops	Risk xATR <= your maximum

Step 4 — Add alerts

• Create an alert on the indicator condition Long Ready or Short Ready.

• Use Once Per Bar or Once Per Bar Close depending on whether you want real-time touch behavior or bar-close confirmation.

• The default message includes ticker, interval, retrace, stop, target, trigger xATR, and risk xATR through plot placeholders.

Step 5 — Manual review

• Open symbols with Ready = 1 and verify market context, spread/liquidity, news, trend, and broader risk conditions.

• Enable Show levels on chart for visual confirmation of retracement, stop, and target.

• Do not assume the reference target or stop is sufficient for your trading plan; treat them as planning levels only.

8. Validation Checklist

• Confirm the script compiles in TradingView Pine Editor v6.

• Test one highly liquid US stock on 5m data and verify the session open resets at 09:30 New York time.

• Compare Use prior-bar ATR on/off to understand how much the opening impulse bar changes the trigger threshold.

• Run the screener on a small watchlist first, then expand to broader watchlists after filters are tuned.

• Review a sample of Long Ready and Short Ready outputs manually before relying on alerts.

9. Limitations and Risk Notice

This script is an educational indicator and screener. It does not provide financial advice, does not place trades, and does not account for all execution risks, spreads, halts, news catalysts, borrow constraints, slippage, or portfolio-level risk. Results can vary by data feed, chart session setting, timeframe, and alert frequency. Validate all logic in TradingView and adapt the settings to your own plan before using it in live decision-making.

Because Pine Screener and alerts rely on TradingView infrastructure and selected settings, changes to watchlists, inputs, timeframe, filters, or script code require rescanning or recreating alerts as applicable. TradingView notes that alert creation stores a snapshot of the script, inputs, and chart context; changes after alert creation do not modify already-created alert instances [5].

Appendix A — Implementation Notes

New-session detection

The original reset pattern relied on an in-session transition: inRTH and not inRTH[1]. That can fail on RTH-only histories where the previous bar is the prior day's final RTH bar. The upgrade also compares the calendar day in the selected session time zone, so the first RTH bar of a new date resets correctly.

ATR handling

By default, the trigger uses prior-bar ATR. This reduces reflexivity: the same bar that creates the impulse does not simultaneously increase the ATR threshold it must exceed. Users can disable this to reproduce the original behavior more closely.

Alert text

The alertcondition() messages are static strings with placeholders. This matches TradingView's alertcondition model, where the message argument is const but placeholders can make emitted alert text dynamic [5][6].

Screener-first plotting

The visible chart plots appear after the hidden screener plots so the first ten plot calls remain the screener contract. This ordering is intentional.

Appendix B — References

[1] TradingView Help Center, "TradingView Pine Screener: key features and requirements."
https://www.tradingview.com/support/solutions/43000742436-tradingview-pine-screener-key-features-and-requirements/

[2] TradingView Pine Script® User Manual, Visuals overview — Pine Screener and plot limitations.
https://www.tradingview.com/pine-script-docs/visuals/overview/

[3] TradingView Pine Script® User Manual, Sessions — time-based sessions and time()/time_close().
https://www.tradingview.com/pine-script-docs/concepts/sessions/

[4] TradingView Pine Script® User Manual, Plots — display.none and plot placeholders.
https://www.tradingview.com/pine-script-docs/visuals/plots/

[5] TradingView Pine Script® FAQ, Alerts — alertcondition behavior and alert snapshots.
https://www.tradingview.com/pine-script-docs/faq/alerts/

[6] TradingView Help Center, "How to use a variable value in alert."
https://www.tradingview.com/support/solutions/43000531021-how-to-use-a-variable-value-in-alert/

[7] TradingView Pine Script® User Manual, Time — time zones and calendar-based functions.
https://www.tradingview.com/pine-script-docs/concepts/time/